Women Techsters Sprint

Capstone Project

Problem Statement

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LEARNING TRACK: Cybersecurity

INDUSTRY TYPE: Education (University)

INDUSTRY NAME: Rexmi State University (RSU)

PROJECT TITLE:

INVESTIGATION OF A RANSOMWARE ATTACK CAUSED BY PHISHING ON A UNIVERSITY LEARNING MANAGEMENT SYSTEM.

1. Background & Context

Higher education institutions and universities increasingly rely on Learning Management Systems (LMS) in the delivery of coursework, assignment, students grading, and communication between students and departments. This makes LMS one of the most critical and popular systems within educational institutions.

The education industry is also one of the most widely attacked sectors using ransomware attacks (an attack that encrypts or locks every file and content to demand money from stakeholders). This is because of the combination of many users within the institution (students and staff) and valuable data stored in these systems.

Phishing attack (which is an email from an attacker disguised as legitimate notice) is still the most common point of entry for this type of cyber-attack. Since it utilizes human weakness in decision making rather than any system vulnerability or weakness, it leaves even well-protected networks exposed. Under such circumstances that RSU was attacked through the most common attack vector in this industry. The attack encrypted the LMS files overnight and, upon the next day, students and staff could no longer access their coursework or communication, but the LMS interface showed a ransom note instead.

2. Problem Statement

The attack posed a bigger challenge regarding the Learning Management System (LMS) for both the staff and students of Rexmi State University (RSU). The system was exposed to an attack chain from phishing to ransomware that caused the total loss of access to learning materials, submission of assignments, grades, and course communication.

This happened after a junior member of the staff opened a link in an email claiming to be an important IT notice, granting the attacker access to the university's network where encryption of the LMS was done overnight. The impact of the attack was experienced by all the RSU staff and students and at a very wrong time during the academic semester, just close to its completion, when assignments and grades are most critical.

This incident did not occur because of any single mistake but because of five particular controls that were lacking or insufficient when they were each required at their exact moment

- No email filtering at the point of entry,
- No SSL/TLS monitoring and inspection of encrypted traffic,

- Unpatched domain controller
- No threat detection tune to send alert when attacker got admin access, and
- Lack of backup segmentation

3. Target Audience

Primary stakeholders: the primary users affected by the attack are RSU students and staff, who rely on the LMS daily. This includes all students who must access the LMS to submit assignments, check their grades, and exchange messages with the lecturers particularly in the period of approaching end of the semester. The lecturers also require the functioning of LMS to grade the students, to manage the educational process, and to communicate with them. The problem faced by the stakeholders is very straightforward and immediate because if the LMS stops working, the academic tasks cannot be accessed anymore.

Secondary stakeholders: IT Department and university administration, whose responsibilities include controlling the situation, bringing the system back to life and communicating with all university stakeholders.

4. Proposed Solution

The proposed solution consists of two parts that are an incident investigation of the RSU cyberattack and the corresponding security fixes covering all vulnerabilities that enabled it.

The first step of the investigation will be the analysis of the attack chain from the beginning (a single user clicking on a phishing link) to the end (encrypting of all files). The reconstruction will take place through the analysis of logs and mapping of MITRE ATT&CK techniques that were used during the attack.

Based on this investigation, the solution suggests the following measures that should be in place which are:

- Email filtering and validation of the email requests enforced at the point of entry, thus preventing any phishing attempts reaching staff's inbox.
- SSL/TLS traffic being inspected to identify command-and-control communications sent through an encrypted channel.

- The vulnerability in the domain controller was fixed to prevent the attacker from achieving instant privilege escalation.
- Thresholds for suspicious user activity being readjusted to detect them in advance prior to obtaining admin access.
- Backup segments of data were separated from the network to ensure that even if an attacker is already present within the network, he cannot delete them.

In this way, the solution will deal with five root causes of the vulnerabilities that caused the incident. This approach can be regarded as fitting RSU's needs, since it does not involve additional resources but specific security control.

5. Learning Track Application

All skills acquired during weeks 2 to 5 of the Cybersecurity track were directly applicable to the RSU scenario.

The skills gained in week 3, concerning threat detection and analysis, have been utilized for developing the attack sequence timeline based on the information found in logs and then mapped to MITRE ATT&CK techniques.

The skills obtained in week 4 regarding vulnerability assessment have been applied for evaluating the impact of the domain controller vulnerability using CVSS to understand whether the vulnerability is low or critical.

The skills developed in week 5, which relate to incident response and governance, have affected the Response Actions and were structured according to the four stages of the NIST Incident Response framework (preparation, detection and analysis, containment/eradication/recovery, and post-incident activity).

The concepts related to networks and access control in week 2 have impacted segmentation and isolation of backups recommendation to restrict further lateral movement in the future and loss of all backups.

6. Conclusion

The project analyzed a ransomware attack caused by phishing against RSU's Learning Management System which resulted in complete encryption of the system files. This project has highlighted five distinct and proven vulnerabilities which made this attack escalate from being a simple click event into an encryption of the entire LMS system in nine hours, and the solution which has been proposed in this project directly addresses all of these vulnerabilities through tangible solutions such as email filtration, inspection of encrypted traffic, patching domain controllers, early threat detection and isolation of backups. This is significant since it not only prevents further attacks but also safeguards the continuity of academic learning for every student and faculty using the LMS at a critical time during the semester. The solution is realistic since it has been created based on the analysis of a specific situation.